PELL'S EQUATION ON ONE PAGE

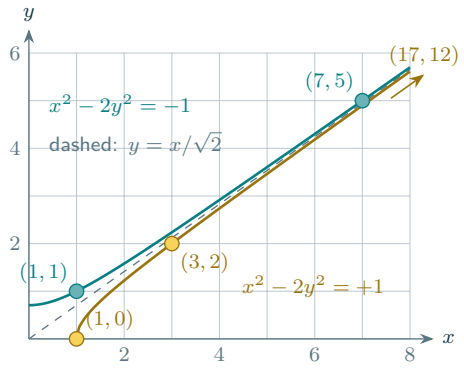
Nearly twice a square

From $3^2 - 2(2^2) = 1$ to the best fractions for $\sqrt{2}$, integer points on a hyperbola, and the norm form $a^2 - 2b^2$.

1 / THE RULE AND THE PICTURE

Squares that miss twice a square by one

Two small coincidences start everything, $3^2 - 2(2^2) = 9 - 8 = 1$ and $17^2 - 2(12^2) = 289 - 288 = 1$, and Pell's equation asks for every such pair, together with its twin that misses by -1: $\boxed{x^2 - 2y^2 = \pm 1}$



x	y	$x^2 - 2y^2$	x/y	error $x/y - \sqrt{2}$
1	1	-1	1	-0.414
3	2	$+1$	1.5	$+0.0858$
7	5	-1	1.4	-0.0142
17	12	$+1$	$1.41\dot{6}$	$+0.00245$
41	29	-1	$1.413793\ldots$	-0.000420
99	70	$+1$	$1.4142857\ldots$	$+0.0000722$
239	169	-1	$1.4142012\ldots$	-0.0000124
577	408	$+1$	$1.4142157\ldots$	$+0.00000212$

The fractions land on alternate sides of $\sqrt{2} = 1.41421356\ldots$, each error about a sixth of the one before, since y grows by $1 + \sqrt{2}$ and so y^2 by $(1 + \sqrt{2})^2 \approx 5.83$.

$$\sqrt{2} \approx \frac{17}{12} \approx \frac{99}{70} \approx \frac{577}{408} \qquad +1 \text{ rows overshoot, } -1 \text{ rows undershoot}$$

2 / WHY THE FRACTIONS ARE SO GOOD

Divide by y^2 and the miss shrinks

Dividing $x^2 = 2y^2 + 1$ by y^2 shows that the square of x/y misses 2 by only $1/y^2$, as in $\left(\frac{17}{12}\right)^2 = \frac{289}{144} = 2 + \frac{1}{144}$. Factorising the difference of two squares turns this into a bound on x/y itself,

$$\frac{x}{y} - \sqrt{2} = \frac{x^2 - 2y^2}{y\,(x + y\sqrt{2})} = \frac{\pm 1}{y\,(x + y\sqrt{2})},$$

so $\left|x/y - \sqrt{2}\right| < 1/(2\sqrt{2}\,y^2)$, far smaller than the $1/(2y)$ that the nearest fraction with an arbitrary denominator y can promise.

$$\frac{17}{12} - \sqrt{2} = \frac{1}{12(17 + 12\sqrt{2})} \approx \frac{1}{408}$$

3 / THE NORM FORM

Multiply, and the norm comes along

In $\mathbb{Z}[\sqrt{2}]$ the number $a + b\sqrt{2}$ times its conjugate $a - b\sqrt{2}$ gives the norm

$$\mathrm{N}(a + b\sqrt{2}) = (a + b\sqrt{2})(a - b\sqrt{2}) = a^2 - 2b^2.$$

The norm is multiplicative, $\mathrm{N}(uv) = \mathrm{N}(u)\,\mathrm{N}(v)$, so powers of the smallest unit generate every solution,

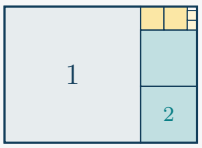
$$(1 + \sqrt{2})^n = x_n + y_n\sqrt{2}, \qquad x_n^2 - 2y_n^2 = (-1)^n,$$

each row of the table following from the last by $(x, y) \mapsto (x + 2y,\ x + y)$. The Gaussian form $a^2 + b^2$ and the Eisenstein form $a^2 - ab + b^2$ belong to the same family; the minus sign here makes the level sets hyperbolas in place of closed curves, so the units never run out.

4 / CUT SQUARES FROM THIS PAGE

The silver rectangle

Euclid's square cutting applied to a $\sqrt{2} \times 1$ rectangle removes one square and from then on always two, because the remainder reproduces itself, $\sqrt{2} - 1 = 1/\big(2 + (\sqrt{2} - 1)\big)$, so $\sqrt{2} = [1; 2, 2, 2, \ldots]$, whose convergents are exactly the table's fractions.



A4 is cut so that halving keeps the shape, hence ratio $\sqrt{2}$, and this page measures 297×210 mm:

$$\frac{297}{210} = \frac{99}{70}$$
$$99^2 - 2(70^2) = 1$$

The golden rectangle sheds one square at every step; the silver one sheds two.

5 / STEPS ALONG THE HYPERBOLA

Each solution is a boost of the last

The step $(x, y) \mapsto (3x + 4y,\ 2x + 3y)$ is multiplication by $3 + 2\sqrt{2}$, a matrix of determinant $9 - 8 = 1$ that slides the hyperbola $x^2 - 2y^2 = 1$ along itself, as a Lorentz boost slides the mass shell $E^2 - p^2 = m^2$. Writing $x = \cosh(n\ell)$ and $y\sqrt{2} = \sinh(n\ell)$, the integer points sit at equal steps of rapidity

$$\ell = \ln(3 + 2\sqrt{2}) = 2\ln(1 + \sqrt{2}) \approx 1.7627,$$

so the rapidity adds while the coordinates multiply, which is why the points race off along the asymptote.

$$x_n + y_n\sqrt{2} = e^{n\ell}, \qquad x_n - y_n\sqrt{2} = e^{-n\ell}$$

The second factor shrinking towards zero is the error of panel 2 seen from the other side.

Convention. Pell's equation $x^2 - dy^2 = 1$ has infinitely many solutions for every positive non-square d (Lagrange, 1768), the smallest for $d = 61$ being $x = 1\,766\,319\,049$; Brahmagupta and Bhaskara II solved such equations centuries before Euler's misattribution to Pell. **Companions:** cutting squares; hyperbola triples; mass shell.

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